Project Stars

A Hybrid Geometric-Relational-Dynamical Model of Thought

Abstract. Project Stars can be formalized as a research graph in which thoughts are modeled simultaneously as attributed objects, nodes in a reviewed and reviewable graph, points in a semantic space, and sources of local activation in a continuous field. The key design principle is epistemic separation: the system may surface candidate relationships, but only reviewed relationships count as confirmed structure. This document presents a paper-style mathematical framing of that model, defines its main objects and operators, distinguishes the underlying thought space from the rendered two-dimensional layout, and explains how candidate inference, structural equilibrium, and activation behavior fit together in one coherent system.

1. Introduction

Project Stars is not merely a visual graph of notes. It is better understood as a knowledge system with explicit mathematical structure. Each thought is represented as an object with semantic content, category membership, provenance, review state, and graph connectivity. Relationships are split into two layers: a confirmed layer of accepted structure and a candidate layer of reviewable hypotheses. The visible graph is then generated by a force-driven embedding whose stable geometry is determined only by confirmed relations, while candidate relations remain inspectable but non-deforming.

This framing matters because it prevents several common confusions. The graph is not identical to the semantic space. The visible layout is not identical to the ontology. And a candidate relation is not the same thing as a confirmed fact. The implementation already suggests these distinctions; the purpose of this document is to make them explicit in mathematical terms.

2. Core Mathematical Object

We define the system as the hybrid object

$$\mathcal{M} = (\Theta, \mathcal{A}, d, R_c, R_p, S, p_t, \phi),$$

where:

- Θ is the set of thoughts,
- $\mathcal{A}$ is the attribute structure carried by each thought,
- d is a multi-component similarity or distance structure,
- R_c is the set of confirmed relations,
- R_p is the set of candidate relations,
- S is the candidate-link scoring functional,

- p_t is the time-dependent visual embedding into $\mathbb{R}^2$,
- ϕ is a continuous activation field over the rendered space.

This is intentionally a coupled model rather than a single familiar mathematical object. It combines an attributed object space, a layered graph, an epistemic distinction between review states, a dynamic layout process, and a field-based activation layer.

3. Thoughts as Structured Objects

Let Θ denote the set of all thoughts represented in the system. Each thought $\theta \in \Theta$ is modeled as

$$\theta = (L, D, e, C, M),$$

with:

- $L(\theta)$: a label string,
- $D(\theta)$: a descriptive text,
- $e(\theta) \in \mathbb{R}^d$: an embedding vector,
- $C(\theta)$: a category or ontology class,
- $M(\theta)$: a metadata bundle.

The metadata bundle is not incidental. It includes references, provenance, evidence class, consensus, review state, source type, and layout state. In other words, a thought is not only a semantic object; it is also an evidential and relational object.

3.1 Attribute Layer

The attribute structure may be regarded as a map

$$\mathcal{A} : \Theta \to \mathcal{L} \times \mathcal{D} \times \mathbb{R}^d \times \mathcal{C} \times \mathcal{M},$$

where $\mathcal{L}, \mathcal{D}, \mathcal{C}, \mathcal{M}$ are the spaces of labels, descriptions, categories, and metadata bundles, respectively. This makes each thought an attributed node rather than an anonymous vertex.

4. Geometry of Thought

The embedding map

$$e : \Theta \to \mathbb{R}^d$$

places each thought into a semantic feature space. A basic semantic dissimilarity can be written as

$$d_{\text{sem}}(\theta_i, \theta_j) = 1 - \cos\big(e(\theta_i), e(\theta_j)\big).$$

However, the system is better described as a *multi-metric* or multi-signal space. Define a composite dissimilarity

$$d(\theta_i, \theta_j) = \sum_{k=1}^{m} \lambda_k d_k(\theta_i, \theta_j), \qquad \lambda_k \geq 0,$$

where the d_k may include semantic, citation, topic, ontology, temporal, or graph-structural distances. This reflects the fact that conceptual relation in Project Stars is not decided by one geometry alone.

5. Layered Relational Structure

A central design choice is the separation between accepted structure and suggested structure.

5.1 Confirmed Relations

The confirmed relation layer is

$$R_c \subseteq \Theta \times \Theta \times \mathcal{T}_R.$$

A tuple $(\theta_i, \theta_j, r) \in R_c$ represents a reviewed or accepted relation of type r. These relations are structurally binding: they define the stable backbone of the graph.

5.2 Candidate Relations

The candidate relation layer is

$$R_p \subseteq \Theta \times \Theta \times \mathcal{T}_R \times [0, 1] \times \mathcal{E}.$$

A tuple in R_p stores a candidate relation together with its score and evidence bundle. Candidate relations are hypotheses. They may be displayed and inspected, but they are not silently promoted to confirmed structure.

5.3 Epistemic Interpretation

This split makes the graph epistemic as well as structural. Confirmed relations encode current commitments. Candidate relations encode current hypotheses. The interface therefore becomes a reviewable knowledge system rather than a static map.

6. Candidate-Link Inference

Candidate links are generated by a deterministic scoring functional rather than by random suggestion. For each thought θ, define reusable signal sets such as:

- $\mathrm{Tok}(\theta)$: token set derived from label and description,
- $\mathrm{Ref}(\theta)$: reference or citation set,
- $\mathrm{Top}(\theta)$: topic set,

- Year(θ): representative temporal signal.

The candidate score between two thoughts is then defined as

$$S(\theta_1, \theta_2) = 0.28\, s_{\text{sem}} + 0.24\, s_{\text{cit}} + 0.18\, s_{\text{top}} + 0.16\, s_{\text{nei}} + 0.09\, s_{\text{ont}} + 0.05\, s_{\text{rec}},$$

where each component is normalized to $[0, 1]$ and denotes, respectively, semantic similarity, citation overlap, shared topics, shared confirmed neighbors, ontology match, and recency weight.

A candidate relation is admitted only when

$$S(\theta_1, \theta_2) \geq \tau,$$

for threshold τ. More precisely, one may define

$$R_p = \{(\theta_i, \theta_j, r, S, E) : S(\theta_i, \theta_j) \geq \tau,\ E \neq \varnothing\}.$$

This formalizes a key principle: candidate relations must have traceable basis.

7. Layout Dynamics

The displayed graph is not the thought space itself. It is a time-dependent embedding of the graph into the plane. Define

$$p_t : \Theta \to \mathbb{R}^2,$$

where $p_t(\theta)$ is the rendered position of thought θ at time t.

If $v_\theta(t) = \frac{d}{dt} p_\theta(t)$ denotes the velocity of node θ, a simplified motion law is

$$m_\theta \ddot{p}_\theta = \sum_{\phi \neq \theta} F_{\text{rep}}(\theta, \phi) + \sum_{(\theta,\phi) \in R_c} F_{\text{spring}}(\theta, \phi) + F_{\text{center}}(\theta) - b\dot{p}_\theta.$$

Here:

- F_{rep} is a pairwise repulsion term,
- F_{spring} is a spring force along confirmed edges,
- F_{center} is a weak inward centering force,
- $b\dot{p}_\theta$ is damping.

7.1 Confirmed-Only Spring Axiom

The key modeling choice is

$$F_{\text{spring}}(\theta, \phi) = 0 \quad \text{if } (\theta, \phi) \notin R_c.$$

This means uncertain hypotheses do not deform the stable structure of the graph. Candidate relations annotate the map, but confirmed relations shape its equilibrium.

7.2 Energy Interpretation

The layout may also be interpreted variationally via an energy of the form

$$E(p) = \sum_{(\theta_i,\theta_j)\in R_c} k_{ij}(\|p_i - p_j\| - \ell_{ij})^2 + \sum_{i<j} \frac{\alpha}{\|p_i - p_j\| + \varepsilon} + \sum_i \mu\|p_i\|^2.$$

Then the visual layout can be viewed as an approximate descent toward a low-energy configuration consistent with confirmed structure.

8. Activation Field

The animated background can be treated as a continuous activation field rather than mere decoration. Let

$$\phi : \mathbb{R}^2 \times \mathbb{R}_{\geq 0} \to \mathbb{R}$$

be a scalar field over the rendered plane. A suitable idealization is a damped driven wave equation,

$$\frac{\partial^2 \phi}{\partial t^2} = c^2 \nabla^2 \phi - \gamma \frac{\partial \phi}{\partial t} + \sum_{\theta\in\Theta} S_\theta(x,t),$$

where $S_\theta(x,t)$ is a localized source term. For a selected thought, one may write

$$S_\theta(x,t) = A_\theta(t) \exp\left(-\frac{\|x - p_\theta(t)\|^2}{2\sigma^2}\right).$$

This expresses the idea that a selected thought acts as a local disturbance in an activation field. The graph gives discrete relational structure; the field gives continuous local influence.

9. Visual Encoding as Epistemic Encoding

Once the mathematical layers are separated, the visual language becomes easier to justify.

- Confirmed edges should encode accepted structural relations.
- Candidate edges should encode hypotheses above threshold.
- Selection and activation effects should encode local attention, not truth status.

This means the interface is not merely styled; it is performing epistemic encoding. Gold or solid treatments may represent accepted structure, while faint dashed or subdued treatments represent reviewable hypotheses. The wave or glow layer should remain distinct from both so that activation is not confused with confirmation.

10. Interpretability Surfaces

The inspector or detail panel is mathematically important because it translates hidden structure into explicit explanation. For a selected thought or relation, the interface should expose:

- evidence class,
- consensus or confidence,
- review state,
- provenance,
- confirmed degree,
- candidate count,
- score components for candidate relations,
- rationale and citations.

Without such surfaces, the graph remains visually suggestive but scientifically weak. With them, the system becomes interpretable and reviewable.

11. Scientific Contribution

The main contribution of Project Stars is not a single algorithm. It is the coupling of several ideas into one coherent, inspectable system:

1. thoughts are treated as structured research objects rather than plain labels,
2. confirmed and candidate relations are separated epistemically and visually,
3. candidate links are generated by explicit signals and preserved with rationale,
4. only confirmed structure determines stable layout,
5. activation is represented as a field process rather than collapsed into graph topology,
6. export can preserve uncertainty, provenance, and review state rather than flattening them away.

This makes the project interpretable as a research system for modeling thought relations rather than a conventional browser graph.

12. Limitations and Extensions

The present formulation remains intentionally practical. It does not yet require a learned embedding model, a full ontology reasoner, or a collaborative review backend. The scoring functional is interpretable but hand-weighted. The field layer is meaningful as an activation model, but it is still partly metaphorical unless tied to stronger behavioral semantics.

Natural extensions include:

- calibrated or learned score weights,
- stronger embedding-based retrieval,
- typed morphisms between thoughts,

- explicit cluster or tightness measures,
- richer ontology structure,
- activation dynamics tied directly to traversal or review events.

A deeper extension would reinterpret confirmed relations as morphisms $r : \theta_i \to \theta_j$ so that paths through the graph acquire compositional meaning. Another would replace the flat semantic space $\mathbb{R}^d$ with a curved manifold so that conceptual distance depends on local structure.

13. Conclusion

Project Stars supports a mathematically serious reading of thought as a structured, relational, and dynamical object. The most important insight is that a thought is not merely content; it also has geometry, evidence, review status, graph role, and field behavior. Once those layers are separated and coupled properly, the system becomes legible as a hybrid model with real explanatory power.

The resulting picture is concise: thoughts live in a multi-signal space, relations exist in confirmed and candidate layers, stable geometry is determined by reviewed structure, and local attention propagates through a continuous activation field. That is the mathematical core of the model.